DATA SCIENCE & AI · MICRO-LECTURE

Who survives the Titanic?

From a night in April 1912 to a supervised learning model

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LEARNING OUTCOMES

By the end of this session, you will be able to…

1

The Data Science cycle

Describe its stages, and place each step of a real project, the Titanic, in the right one.

2

Supervised learning

Explain how a model learns from labelled examples, and judge it fairly on data it has never seen.

The Data Science cycle

Six stages, one loop. Each one has a job in the Titanic project.



From a tragedy to a table

≈2,220
people on board, 14–15 April 1912

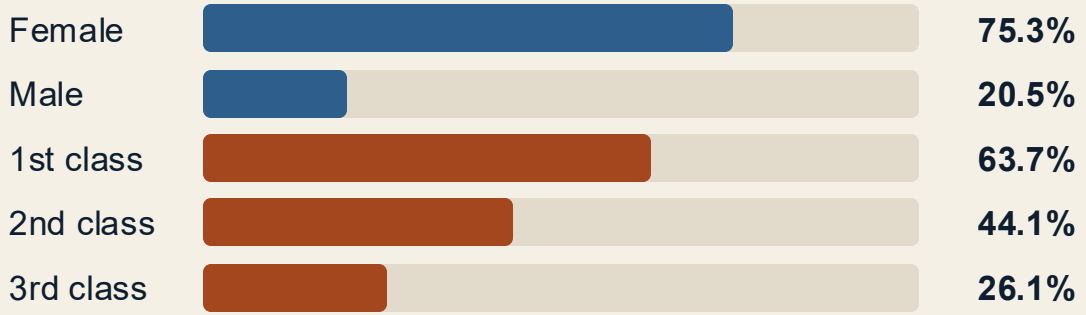
≈1,500
died in the sinking

1,178
lifeboat seats

Every passenger becomes a row

Variable	Values	Role
Survived	Yes / No	Label y (to predict)
Class	1st, 2nd, 3rd	Feature x_1
Sex	female, male	Feature x_2
Age	0.2 – 80 years	Feature x_3

Explore: survival rate



Data: carData::TitanicSurvival, 1,309 passengers; 263 with unknown age removed, leaving n = 1,046 (overall survival 40.8%). Crew not included.

BACK TO THE CAR WINDOW

What does the model predict for Rose and Jack?

Rose

1st class · female · 17

chance of survival

Jack

3rd class · male · 20

chance of survival

A model gives **probabilities, not destinies**. It learned the patterns of 1912 (the lifeboat rule, the deck you slept on), not their causes, and it knows nothing that isn't in the table.

Your turn: open the Titanic Shiny app, then split, fit and predict your own chances

BACK TO THE CAR WINDOW

What does the model predict for Rose and Jack?

Rose
1st class · female · 17

94%
chance of survival

Jack
3rd class · male · 20

12%
chance of survival

A model gives **probabilities, not destinies**. It learned the patterns of 1912 (the lifeboat rule, the deck you slept on), not their causes, and it knows nothing that isn't in the table.

Your turn: open the Titanic Shiny app, then split, fit and predict your own chances